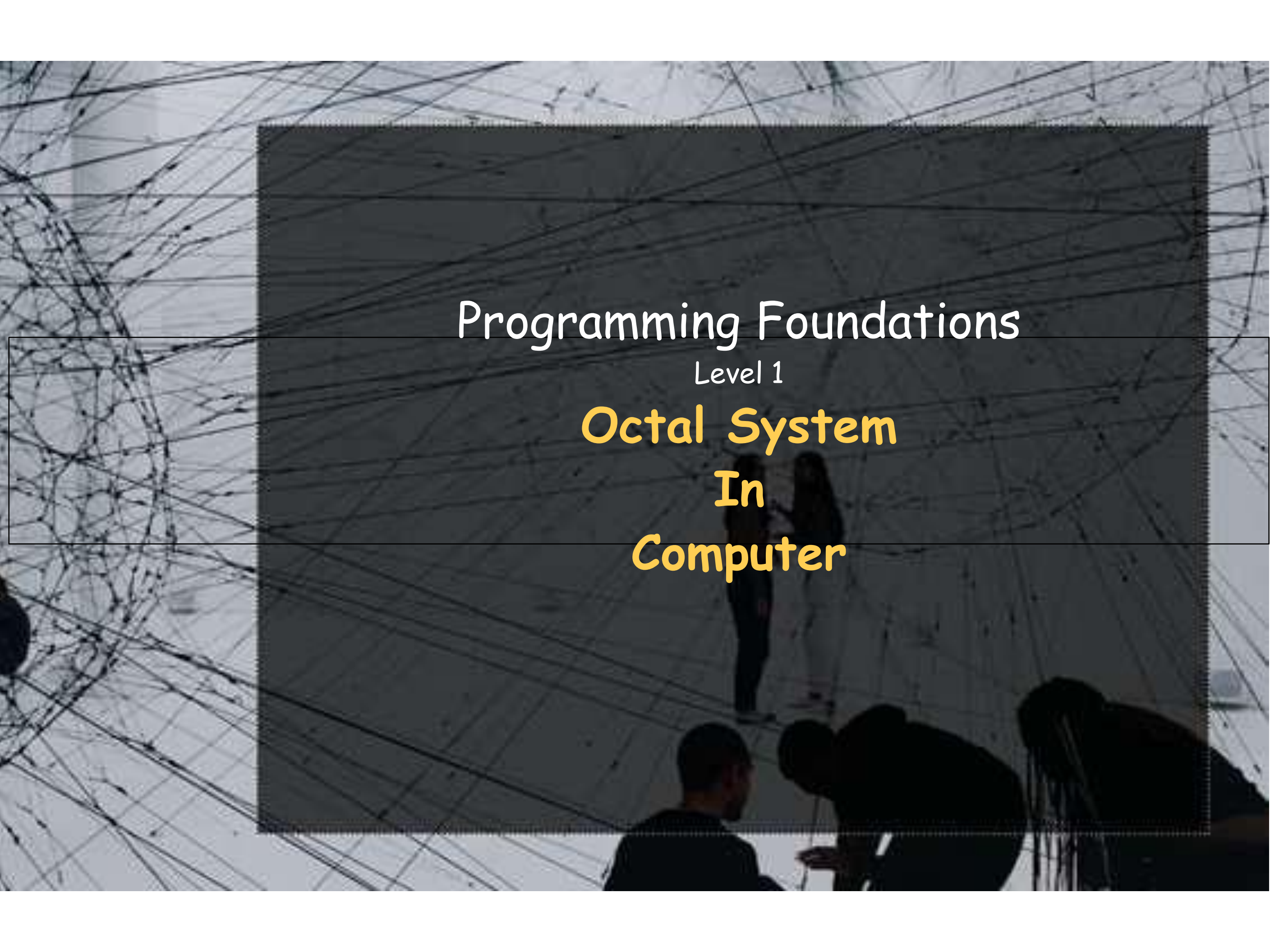




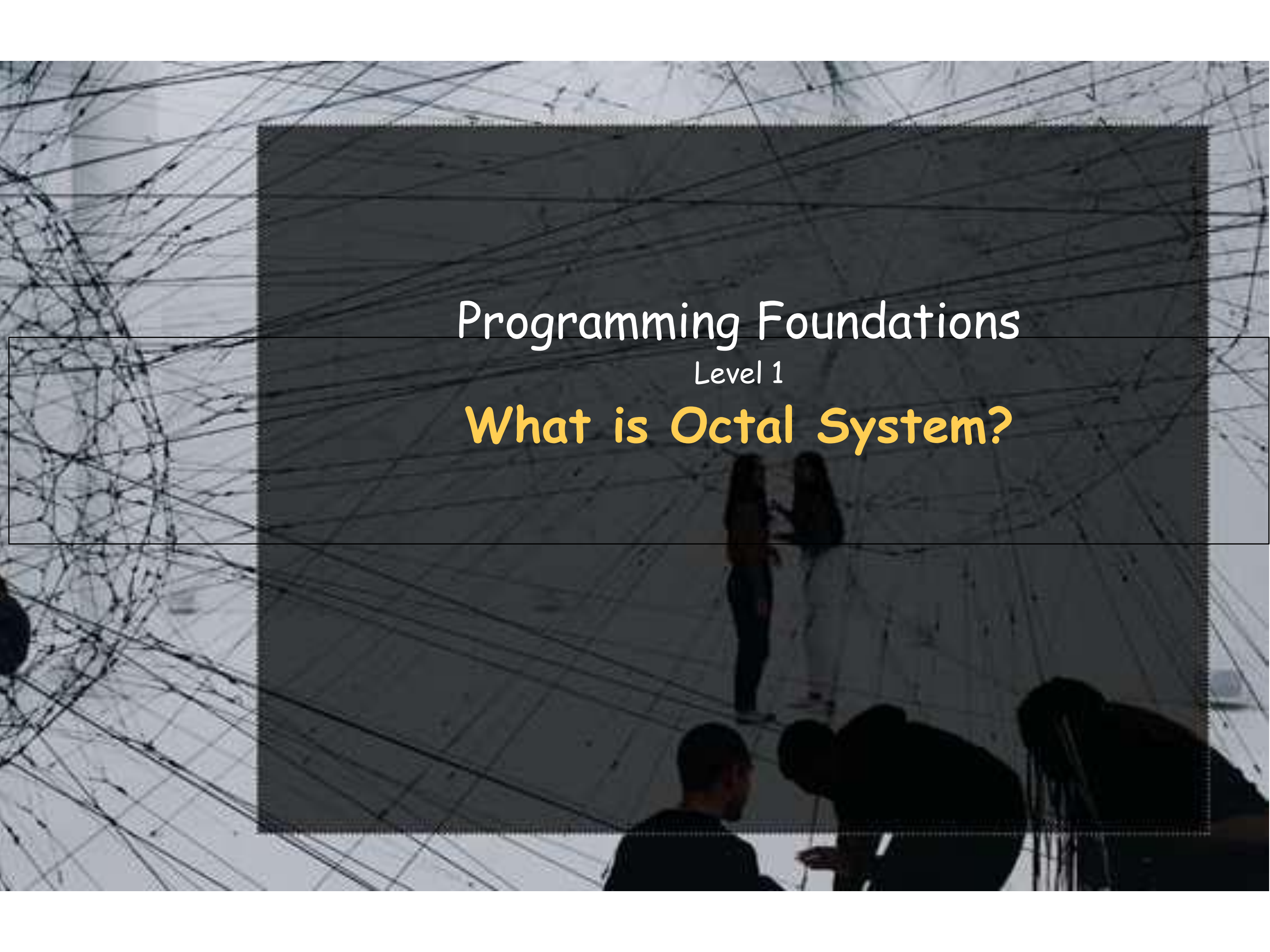
Programming Foundations Level 1

The background of the slide features a complex, abstract pattern of thin, dark lines crisscrossing over a lighter, textured surface. In the lower portion of the image, there are dark silhouettes of several people, some standing and some appearing to be in a discussion or collaborative work, which adds a human element to the technical theme.

Programming Foundations

Level 1

Octal System In Computer



Programming Foundations

Level 1

What is Octal System?

Octal Number System (Base-8): Concepts & Conversions

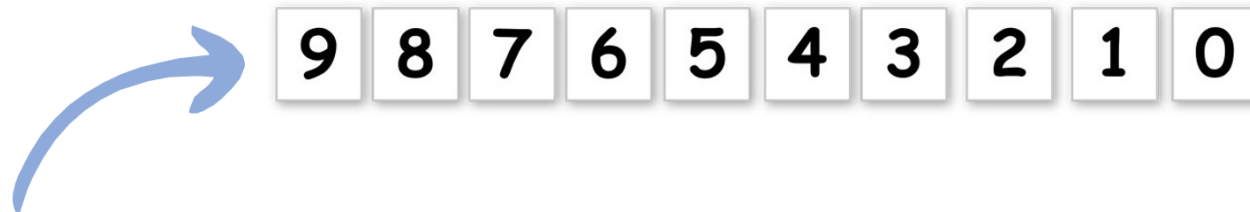
The **Octal System (base-8)** uses digits 0-7. Like hexadecimal, octal provides a **human-friendly view** of binary: each **octal digit = 3 bits**, so long binary strings become shorter and easier to read (e.g., 11010100 → octal **212**).

Historically, octal was popular when machines used **word sizes in multiples of 3 bits** or **6-bit bytes**; modern systems use **nibbles (4 bits)**, so hexadecimal is more common today.

Counting Systems (Context)

Decimal (base-10), Binary (base-2), Hexadecimal (base-16), and Octal (base-8) appear together in computing foundations.

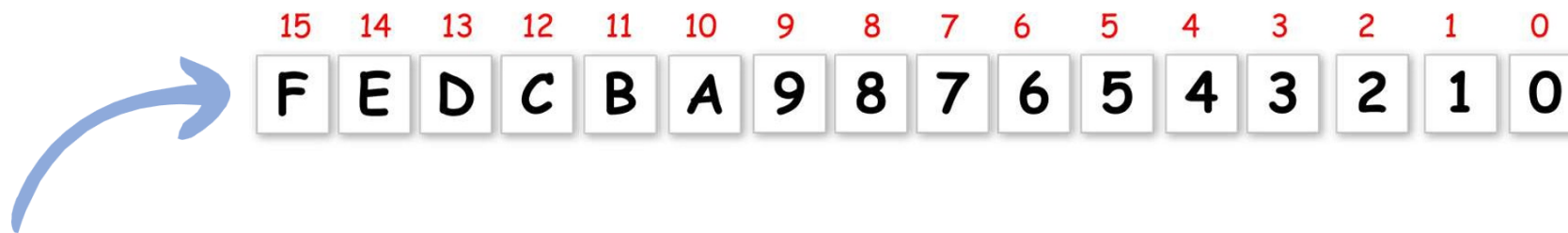
Counting Systems?



Decimal System (Base 10)



Binary System (Base 2)



Hexadecimal System (Base 16)



Octal System (Base 8)

What Is Octal?

The **Octal Numbering System** is very similar in principle to the previous hexadecimal numbering system except that in Octal,

Each octal digit represents 3 bits, so a 6-bit byte is two octal digits

Digits: 0, 1, 2, 3, 4, 5, 6, 7

Place values (powers of 8):

$8^0=1$, $8^1=8$, $8^2=64$, $8^3 = 512$...etc

Binary pairing: 1 octal digit $\leftrightarrow$ 3 bits (e.g., 7 $\leftrightarrow$ 111).

A 6-bit value is exactly two octal digits.

ASCII Table

dec	hex	oct	char	dec	hex	oct	char	dec	hex	oct	char	dec	hex	oct	char
0	0	000	NULL	32	20	040	space	64	40	100	@	96	60	140	`
1	1	001	SOH	33	21	041	!	65	41	101	A	97	61	141	a
2	2	002	STX	34	22	042	"	66	42	102	B	98	62	142	b
3	3	003	ETX	35	23	043	#	67	43	103	C	99	63	143	c
4	4	004	EOT	36	24	044	\$	68	44	104	D	100	64	144	d
5	5	005	ENQ	37	25	045	%	69	45	105	E	101	65	145	e
6	6	006	ACK	38	26	046	&	70	46	106	F	102	66	146	f
7	7	007	BEL	39	27	047	'	71	47	107	G	103	67	147	g
8	8	010	BS	40	28	050	(	72	48	110	H	104	68	150	h
9	9	011	TAB	41	29	051	)	73	49	111	I	105	69	151	i
10	a	012	LF	42	2a	052	*	74	4a	112	J	106	6a	152	j
11	b	013	VT	43	2b	053	+	75	4b	113	K	107	6b	153	k
12	c	014	FF	44	2c	054	,	76	4c	114	L	108	6c	154	l
13	d	015	CR	45	2d	055	-	77	4d	115	M	109	6d	155	m
14	e	016	SO	46	2e	056	.	78	4e	116	N	110	6e	156	n
15	f	017	SI	47	2f	057	/	79	4f	117	O	111	6f	157	o
16	10	020	DLE	48	30	060	0	80	50	120	P	112	70	160	p
17	11	021	DC1	49	31	061	1	81	51	121	Q	113	71	161	q
18	12	022	DC2	50	32	062	2	82	52	122	R	114	72	162	r
19	13	023	DC3	51	33	063	3	83	53	123	S	115	73	163	s
20	14	024	DC4	52	34	064	4	84	54	124	T	116	74	164	t
21	15	025	NAK	53	35	065	5	85	55	125	U	117	75	165	u
22	16	026	SYN	54	36	066	6	86	56	126	V	118	76	166	v
23	17	027	ETB	55	37	067	7	87	57	127	W	119	77	167	w
24	18	030	CAN	56	38	070	8	88	58	130	X	120	78	170	x
25	19	031	EM	57	39	071	9	89	59	131	Y	121	79	171	y
26	1a	032	SUB	58	3a	072	:	90	5a	132	Z	122	7a	172	z
27	1b	033	ESC	59	3b	073	;	91	5b	133	[	123	7b	173	{
28	1c	034	FS	60	3c	074	<	92	5c	134	\	124	7c	174	
29	1d	035	GS	61	3d	075	=	93	5d	135	]	125	7d	175	}
30	1e	036	RS	62	3e	076	>	94	5e	136	^	126	7e	176	~
31	1f	037	US	63	3f	077	?	95	5f	137	_	127	7f	177	DEL

Why Octal (then & now)?

Both Hexadecimal and Octal Systems Provides a
human-friendly representation

11010100 → D4 Hexa

11010100 → 212 Octal

It's hard for human to read binary!

What does this mean?

01001001 00100000 01001100 01101111 01110110
01100101
00100000 01011001 01101111 01110101 00100001

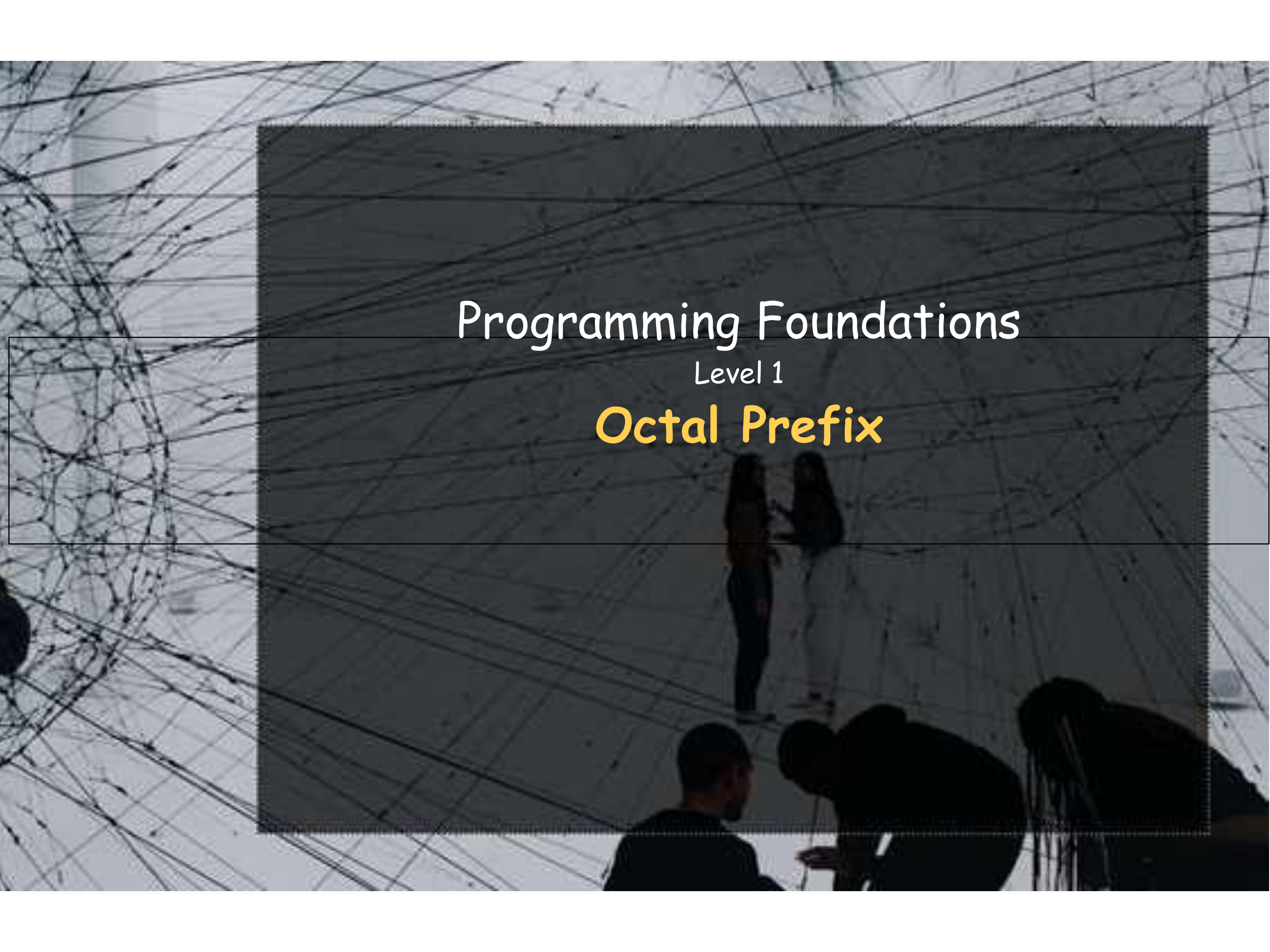


I Love You!



Why Octal (then & now)?

- **Then:** Early computers often used multiples of 3 bits (e.g., 6-bit bytes) → octal fit perfectly.
- **Now:** Most modern machines use nibbles (4 bits) → hexadecimal aligns better, so octal use has **declined**. Still, octal remains helpful for compact binary reading.

The background of the slide features a complex, abstract pattern of thin, dark lines crisscrossing over a lighter, textured surface. In the lower portion of the image, the silhouettes of several people are visible, appearing to be in a collaborative setting, possibly a meeting or a workshop. The overall aesthetic is technical and modern.

Programming Foundations

Level 1

Octal Prefix

Prefix:

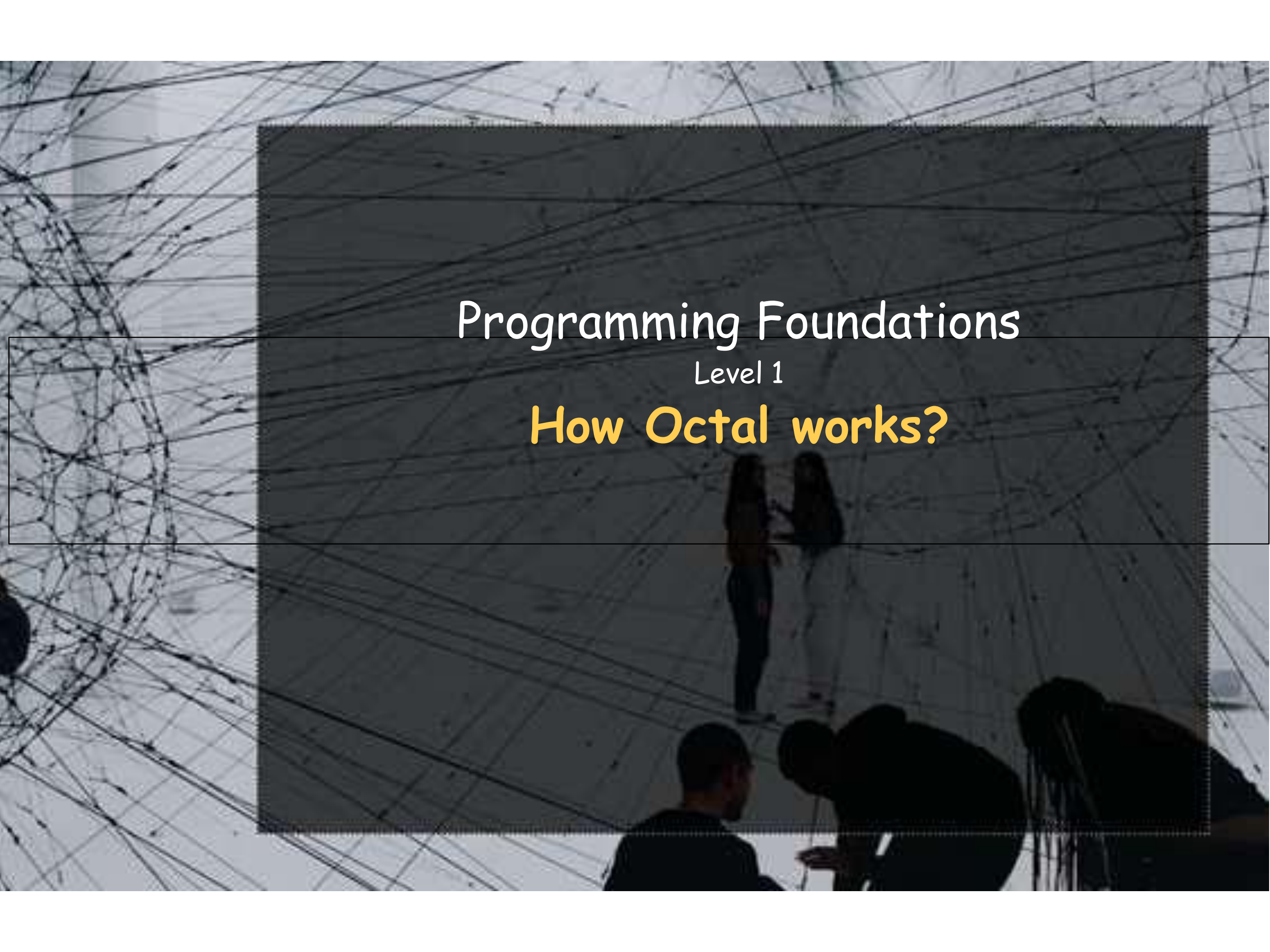
Hexa Prefix:

<u>Technology/Language</u>	<u>Prefix</u>	<u>Example</u>
HTML & CSS	#Code	#FFFFFF
C,C++,Java..etc	0x Code	0x725
XML	&#Code	&#C2A4
Unicode	U+Code	U+C2A4

Octal Prefix:

0oCode → 0o725

- Hex examples (for contrast): #FFFFFF, 0x725, &#C2A4, U+C2A4.
- Octal prefix (programming): 0o (letter o).
Example: 0o725 is octal 725.



Programming Foundations

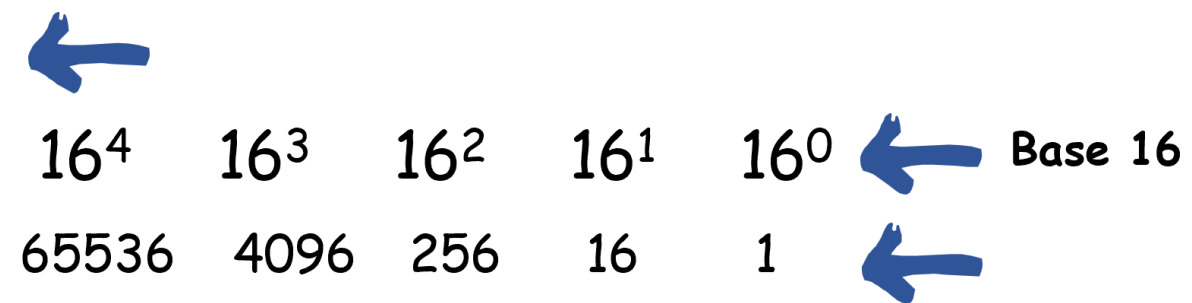
Level 1

How Octal works?

Remember that binary was base 2.

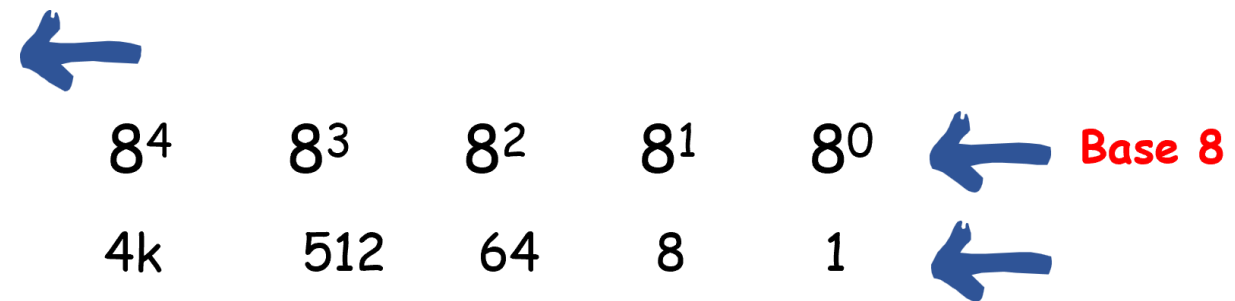
2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0	 Base 2
128	64	32	16	8	4	2	1	 Doubling

Hexadecimal is base 16.



	16^4	16^3	16^2	16^1	16^0		Base 16
	65536	4096	256	16	1		

Octal System is base 8



A diagram illustrating the octal system's base 8. It shows a sequence of powers of 8 from 8^4 down to 8^0 , with their corresponding decimal values below them. Blue arrows point from the right towards the powers, and a red label 'Base 8' is positioned to the right of the arrows.

8^4	8^3	8^2	8^1	8^0	← Base 8
4k	512	64	8	1	←



Programming Foundations

Level 1

Convert Decimal to Octal



Conversions

conversion from decimal to octal and binary to octal

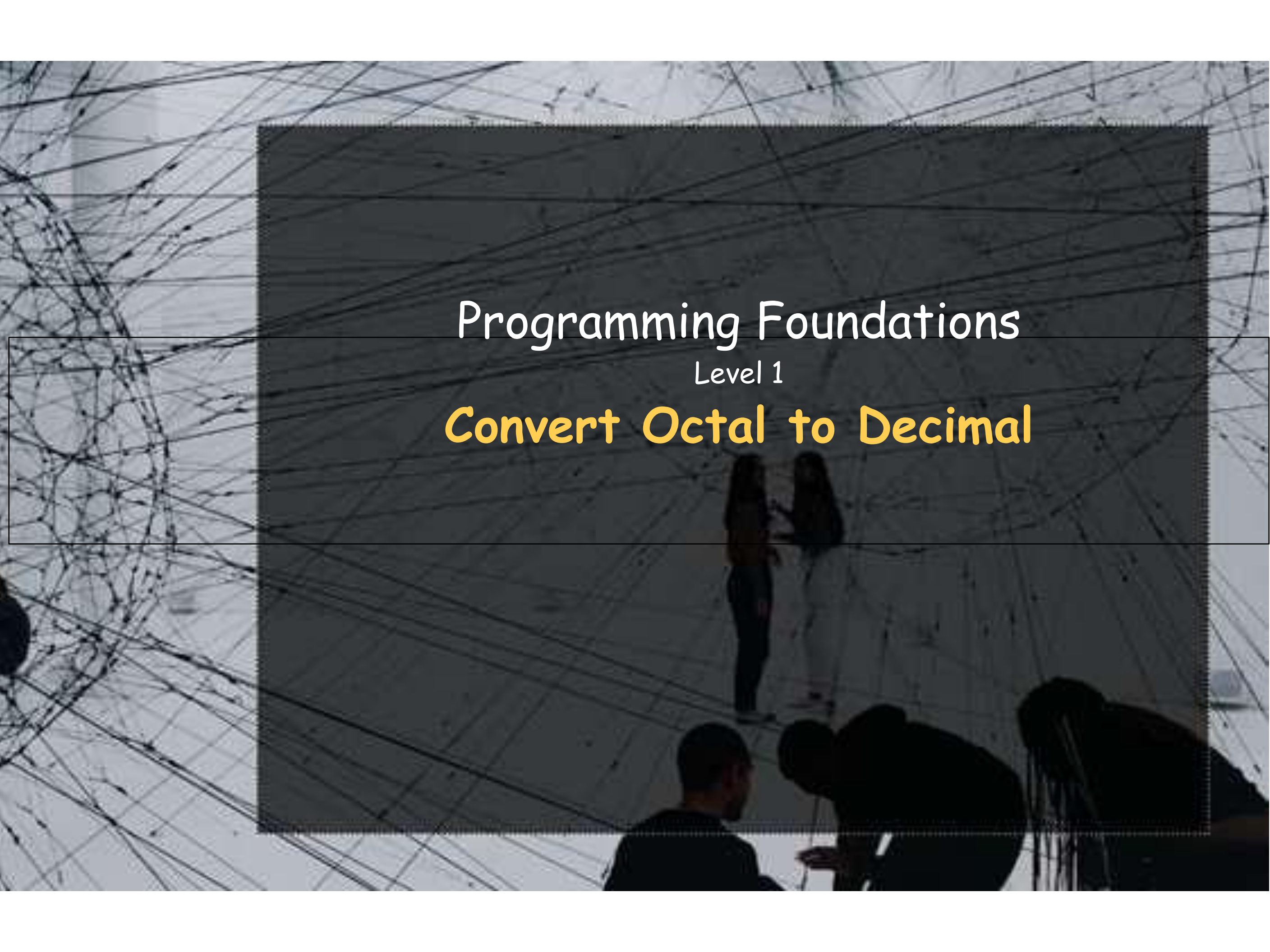
Follows the same pattern as we have seen
previously for hexadecimal

A) Decimal $\rightarrow$ Octal (long division by 8)

Convert 469 to Octal

<u>Number / 8</u>	<u>Result</u>	<u>Integer</u>	<u>Fraction</u>	<u>Remainder</u>		<u>Octal</u>
469 / 8 =	58.625	58	0.625	8 $\times$ 0.625 = 5	$\rightarrow$	5
58 / 8 =	7.25	7	0.25	8 $\times$ 0.25 = 2	$\rightarrow$	2
7 / 8 =	--	--	--	7	$\rightarrow$	7

$469_{10} \rightarrow 725_8$



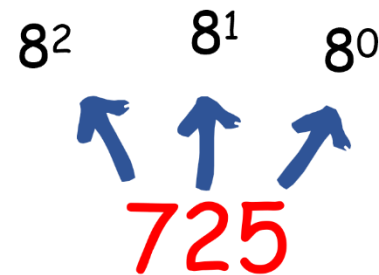
Programming Foundations

Level 1

Convert Octal to Decimal

B) Octal $\rightarrow$ Decimal (expand by power of 8)

Convert Octal **725** to Decimal

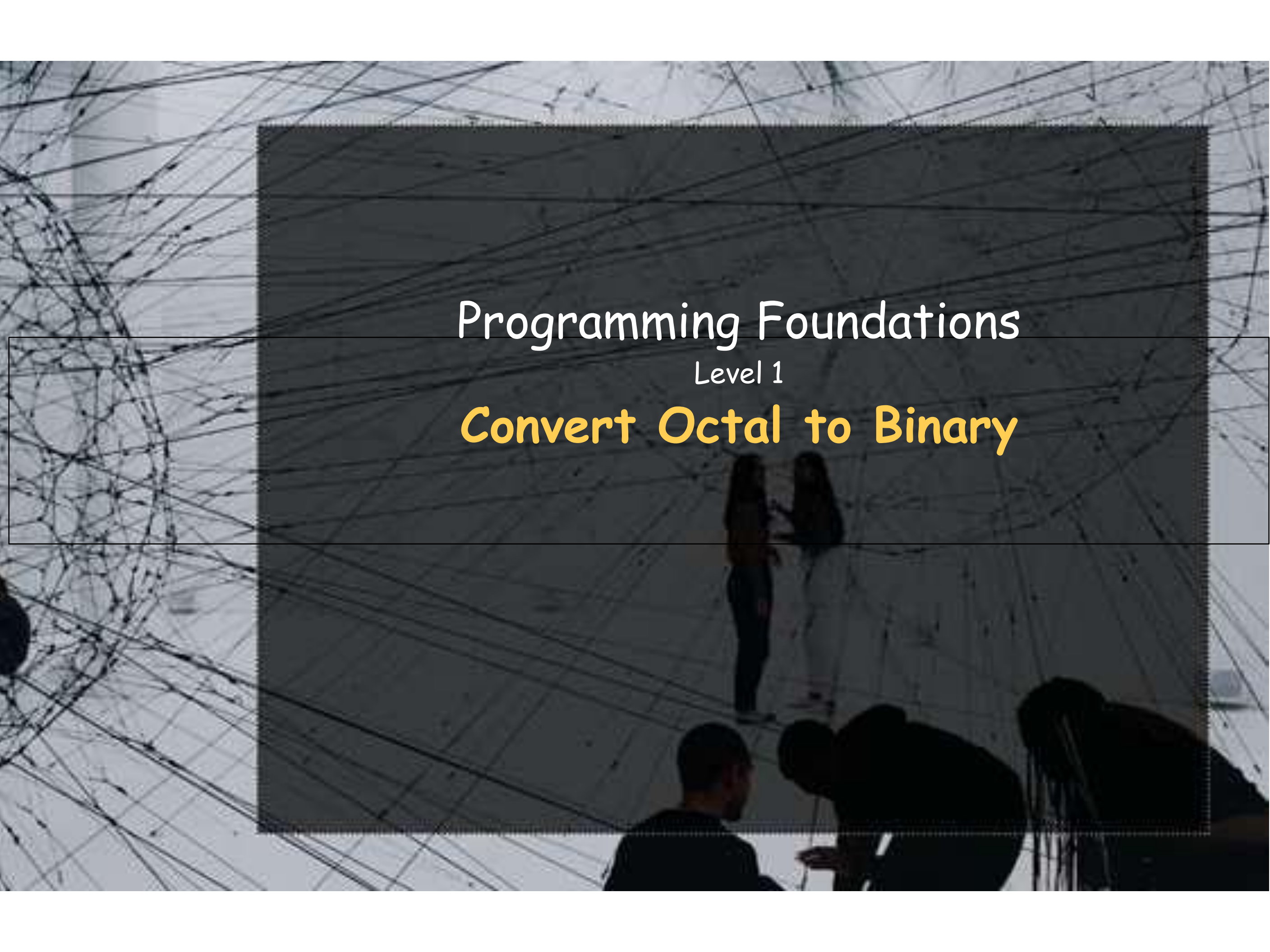


$$5 \times 8^0 = 5 \times 1 = 5 +$$

$$2 \times 8^1 = 2 \times 8 = 16 +$$

$$7 \times 8^2 = 7 \times 64 = 448$$

$$\textbf{469}$$



Programming Foundations

Level 1

Convert Octal to Binary

C) Octal → Binary

How to Convert Octal to Binary?

Way 1:

Two steps:

1. Convert octal to decimal.
2. Convert decimal to binary.

That's it.

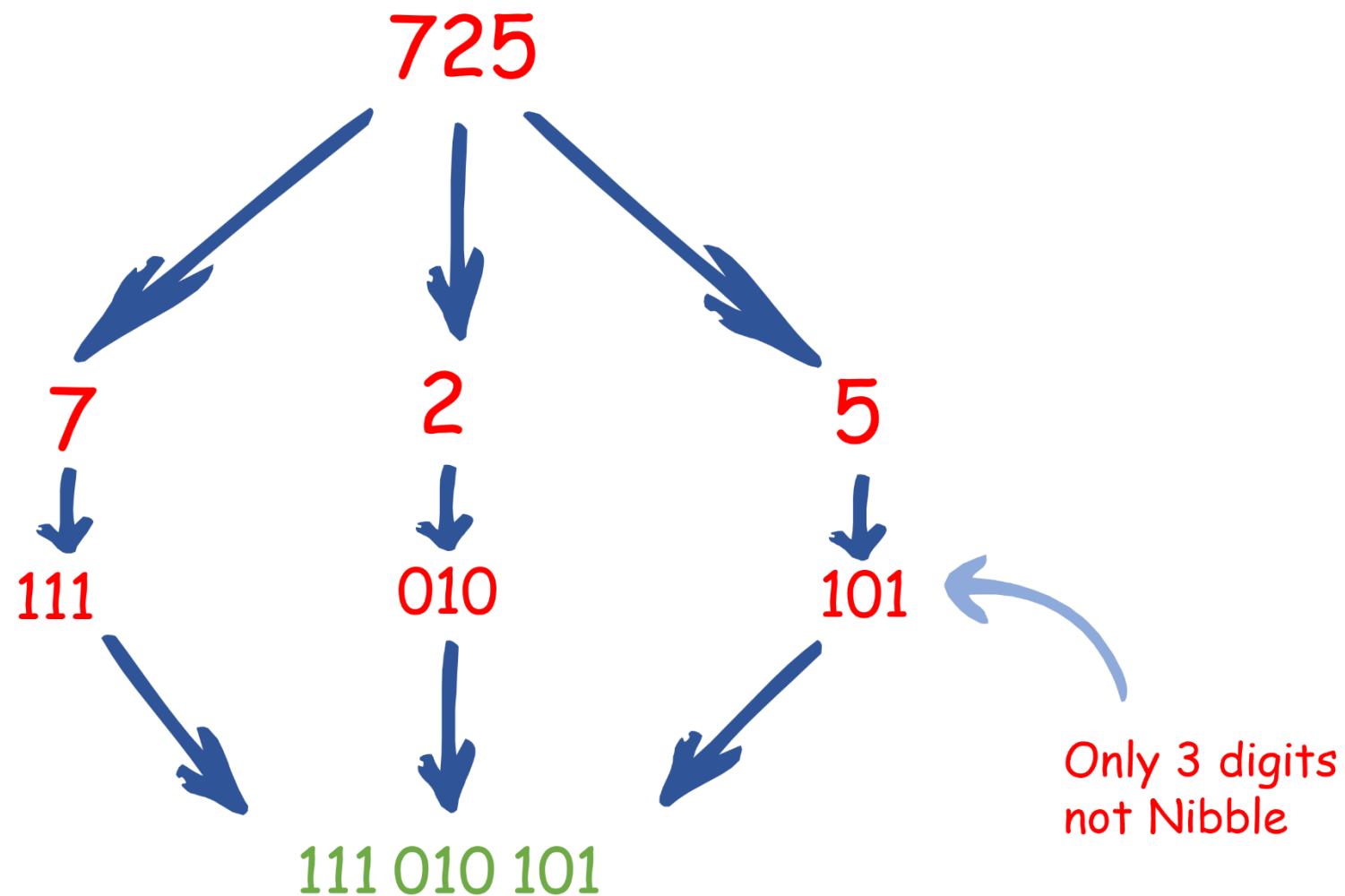
How to Convert Octal to Binary?

Way 2:

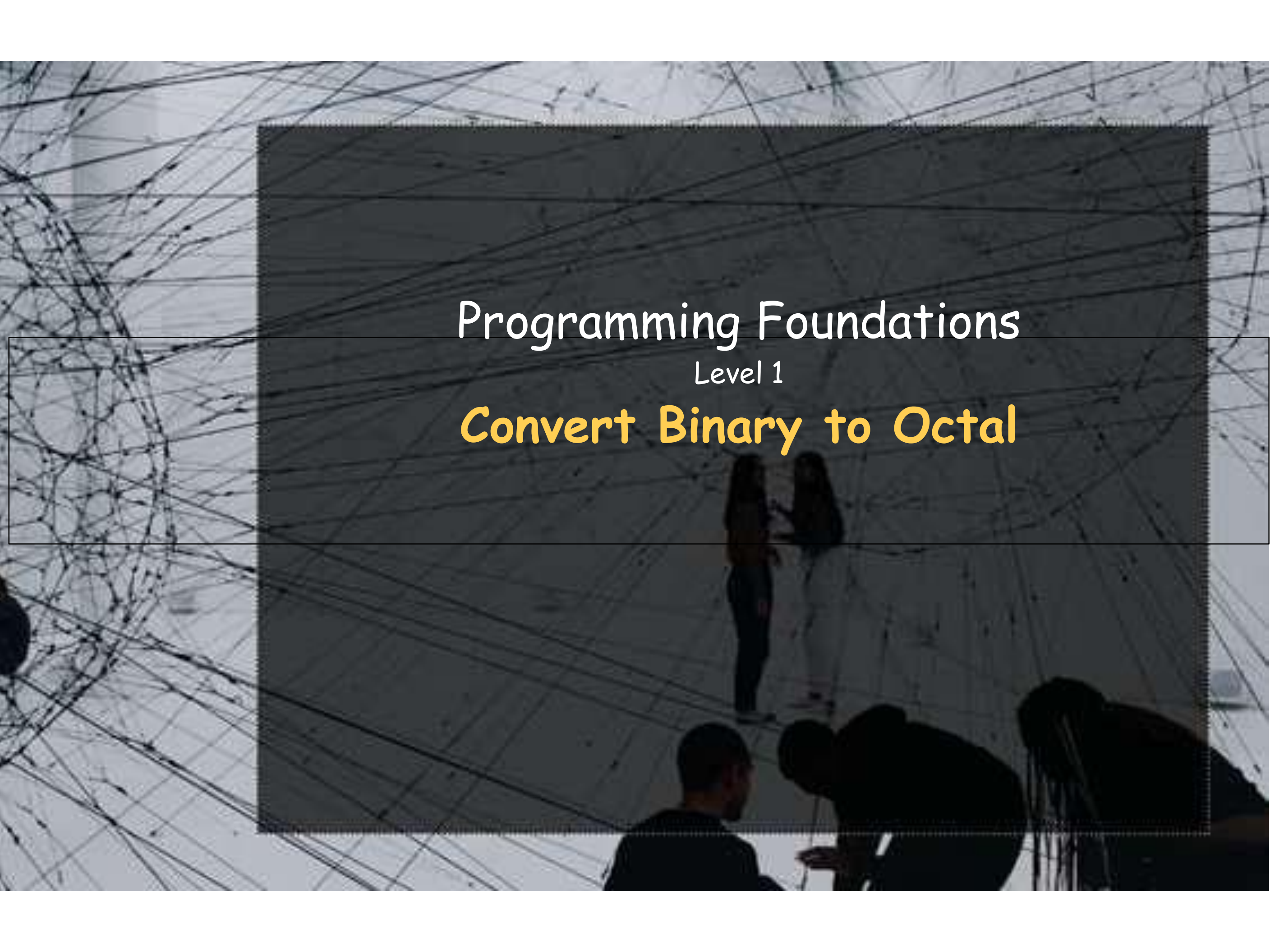
Direct Conversion.

Direct Conversion:

Convert Octal **725** to Binary



7 | 2 | 5 → 111 | 010 | 101 → **111010101** (group as 3-bit blocks).
(You may pad left with zeros for alignment.)



Programming Foundations

Level 1

Convert Binary to Octal

How to Convert Binary to Octal?

Way 1:

Two steps:

1. Convert Binary to Decimal.
2. Convert Decimal to Octal.

That's it.

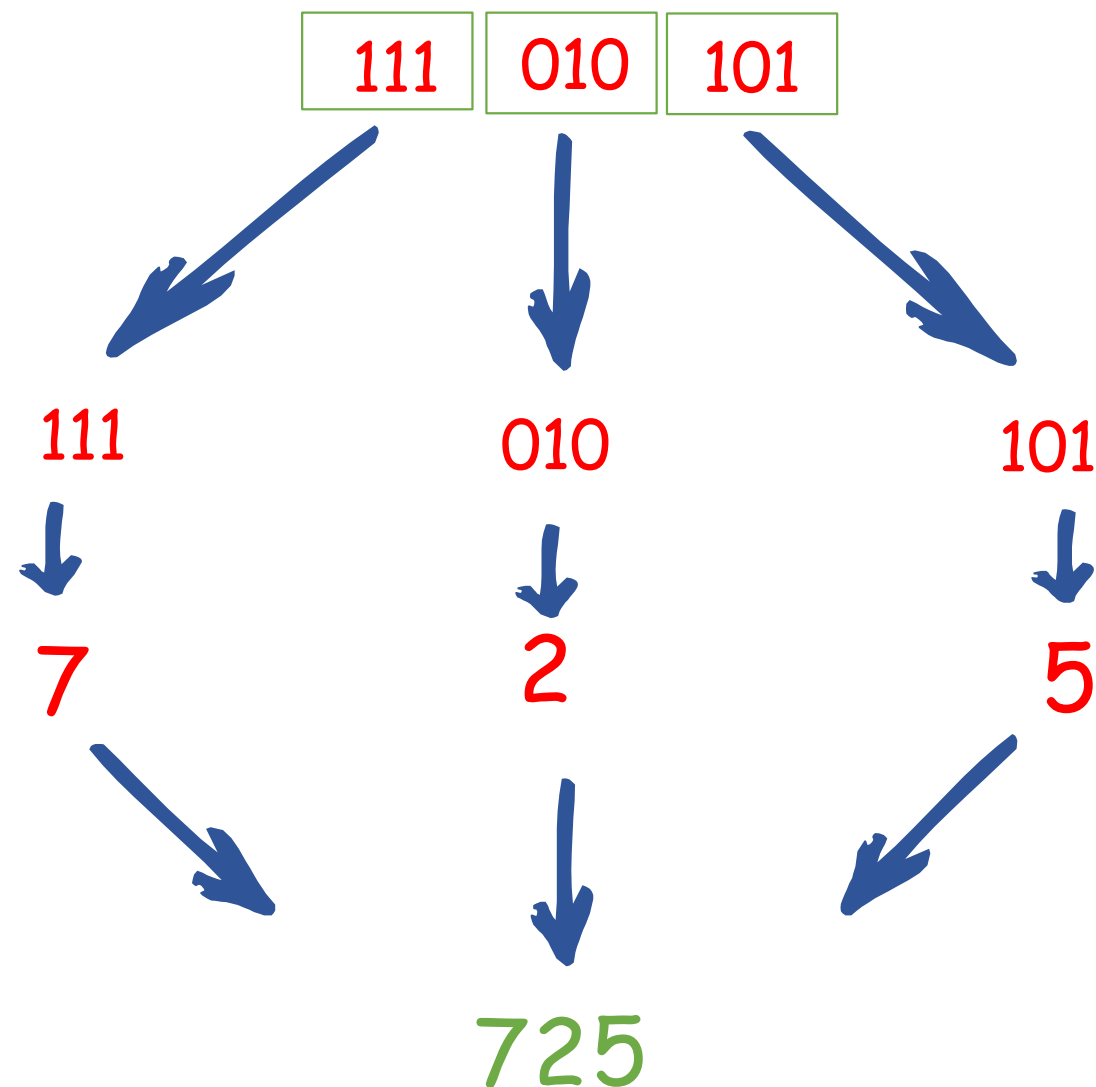
How to Convert Binary to Octal?

Way 2:

Direct Conversion.

Direct Conversion:

Convert **000111010101** to Octal



Group by 3 from right: 000 | 111 | 010 | 101 → 0 7 2 5 → **725** (drop leading 0).

Interconnection

- Octal digit = 3 bits → compact view for older 3-bit-multiple word sizes.
- Hex digit = 4 bits → better match for modern nibble-based architectures.
- Two conversion ways exist for Octal↔Binary; **direct 3-bit grouping** is the quickest in practice.

By knowing where octal shines (and where hex does), you can read binary data efficiently in any context. 🚀

Activity

Instructions:

1 Octal $\rightarrow$ Decimal

a) 100_8

b) 512_8

2 Decimal $\rightarrow$ Octal

a) 64_{10}

b) 330_{10}

3 Octal $\rightarrow$ Binary (direct, 3-bit mapping)

a) 100_8

b) 512_8

4 Binary $\rightarrow$ Octal (direct, group by 3 from right)

a) $0100\ 0000_2$

b) $0001\ 0100\ 1010_2$

Activity Solution

Solution:

1 Octal $\rightarrow$ Decimal

a) $100_8 = 64_{10}$

b) $512_8 = 330_{10}$

2 Decimal $\rightarrow$ Octal

a) $64_{10} = 100_8$

b) $330_{10} = 512_8$

3 Octal $\rightarrow$ Binary (direct)

a) $100_8 \rightarrow 1|0|0 \rightarrow 001\ 000\ 000 \rightarrow 0100\ 0000_2$ (padded to a full byte)

b) $512_8 \rightarrow 5|1|2 \rightarrow 101\ 001\ 010 \rightarrow 0001\ 0100\ 1010_2$ (padded to 12 bits)

4 Binary $\rightarrow$ Octal (direct)

a) $0100\ 0000_2 \rightarrow$ group $010|000|000 \rightarrow 4\ 0\ 0 \rightarrow 100_8$

b) $0001\ 0100\ 1010_2 \rightarrow$ group $000|101|001|010 \rightarrow 0\ 5\ 1\ 2 \rightarrow 512_8$ (drop leading 0)

🏁 Conclusion

- ✓ Octal (base-8) compresses binary using 3-bit groupings per digit.
- ✓ Use the direct methods for speed: *Octal* ↔ *Binary* via 3-bit mapping, *Binary* ↔ *Octal* by grouping in 3s from the right.
- ✓ While **hex** is more common today, octal remains an important historical and practical lens for reading binary succinctly.

Thank you 😊